

Project Documentation

Phase 7 of 8 - Project Documentation

Project: OptiCrop: Smart Agricultural Production Optimization Engine

Project Overview

OptiCrop is a machine learning web application that recommends the most suitable crop for a given set of soil and climate conditions - Nitrogen, Phosphorous, Potassium, temperature, humidity, pH, and rainfall - using a Logistic Regression classifier trained on 22 crop types, complemented by K-Means clustering for exploratory analysis of environmental groupings.

Setup Instructions

- Install Python 3.9+ and the required libraries: numpy, pandas, matplotlib, seaborn, scikit-learn, flask.
- Open Training/OptiCrop_Model_Building.ipynb in Jupyter Notebook and run all cells to reproduce data exploration, preprocessing, clustering, model training, and to regenerate Flask/model.pkl.
- From the Flask/ directory, run 'python app.py' to start the development server.
- Open <http://127.0.0.1:5000> in a browser to use the application.

Technology Stack

Layer	Technology
Data Processing	Pandas, NumPy
Visualization	Matplotlib, Seaborn
Machine Learning	Scikit-learn (KMeans, LogisticRegression)
Model Serialization	Pickle
Web Framework	Flask
Frontend	HTML5, CSS3 (Jinja2 templating)

References

- Kaggle Dataset - Smart Agricultural Production Optimizing Engine
- Scikit-learn documentation - <https://scikit-learn.org/stable/>
- Flask documentation - <https://flask.palletsprojects.com/>

- Seaborn documentation - <https://seaborn.pydata.org/>